C PROGRAMMING • LECTURE 03

Operators in C

Arithmetic • Relational • Logical • Assignment • Increment/Decrement • Conditional • Bitwise

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Batch • CSE-60

```
int a = 10, b = 3;
int sum = a + b;
int r = a % b;
a++;
```

What is an Operator?

A symbol that tells C to perform an operation on one or more values.

Operands

Values or variables on which an operator works.

```
a + b
↑   ↑
operands
```

Operator

The symbol describing the operation.

```
a   +   b

+  →  addition
```

Arithmetic Operators

Used for mathematical calculations.

+ Addition

```
a + b
```

- Subtraction

```
a - b
```

* Multiplication

```
a * b
```

/ Division

```
a / b
```

% Remainder

```
a % b
```

% gives the remainder and is used with integer operands.

Integer Division vs Fractional Division

```
int a = 7, b = 2;

printf("%d", a / b);          // 3
printf("%.1f", (float)a / b); // 3.5
```

Remember

- int / int → integer result
- Use a cast for a fractional result
- 7 / 2 = 3, not 3.5

Relational Operators

Used to compare two values. The result is true or false.

`==`	equal to
`!=`	not equal
`>`	greater than
`<`	less than
`>=`	greater/equal
`<=`	less/equal

```
int a = 10, b = 20;
printf("%d", a < b);  // 1
printf("%d", a == b); // 0
```

Logical Operators

Combine or reverse conditions.

&& AND

both conditions must be true

```
age >= 18 && age <= 60
```

|| OR

at least one condition is true

```
marks >= 80 || grade == 'A'
```

! NOT

reverses true ↔ false

```
!(x > 5)
```

Assignment Operators

Store or update a value in a variable.

`=`	`a = 5`
`+=`	`a += 2`
`-=`	`a -= 2`
`*=`	`a *= 2`
`/=`	`a /= 2`
`%=`	`a %= 2`

Example: a += 2 is the same as a = a + 2.

++ and --

Increase or decrease a variable by exactly 1.

```
int x = 5;

x++;   // x becomes 6
x--;   // x becomes 5
```

Two forms

`++x` → **prefix**

`x++` → **postfix**

Both increase x by 1; the difference matters inside a larger expression.

The Important Difference

```
int x = 5, a, b;

a = x++;  // a = 5, x = 6
x = 5;
b = ++x;  // b = 6, x = 6
```

Easy memory trick

POST = use first, then increase.

PRE = increase first, then use.

Ternary / Conditional Operator

A compact alternative to a simple if-else.

```
int a = 10, b = 20;

int max = (a > b) ? a : b;
printf("%d", max);
```

Pattern

`condition ? value1 : value2`

true → value1
false → value2

Bitwise Operators

Operate directly on the binary bits of integer values.

`&`	AND
`|`	OR
`^`	XOR
`~`	NOT
`<<`	Left shift
`>>`	Right shift

Focus first on &, |, ^, << and >>; practice ~ after the basics.

See the Bits

```
int a = 5;   // 0101
int b = 3;   // 0011

a & b = 0001 = 1
a | b = 0111 = 7
a ^ b = 0110 = 6
```

XOR rule

Same bits → 0
Different bits → 1

`5 ^ 3 = 6`

Which Operation Happens First?

Use parentheses when the intended order is important.

1	`()`	Parentheses
2	`++ -- !`	Unary
3	`* / %`	Multiply / divide / remainder
4	`+ -`	Add / subtract
5	`< <= > >=`	Relational
6	`== !=`	Equality
7	`&&`	Logical AND
8	`||`	Logical OR
9	`= += -= ...`	Assignment

Tip

When unsure, use parentheses.

```
a + b * c
≠
(a + b) * c
```

Predict the output

Try first. Then verify by compiling.

```
int a = 10, b = 3;

printf("%d\n", a + b);
printf("%d\n", a / b);
printf("%d\n", a % b);
```

Your answer

Write it here:

_____

Answer: 13, 3, 1

Relational + logical

Try first. Then verify by compiling.

```
int a = 5, b = 10;

printf("%d\n", a < b);
printf("%d\n", a > b);
printf("%d\n", a < b && b > 8);
printf("%d\n", a == 5 || b == 5);
```

Your answer

Write it here:

_____

Answer: 1, 0, 1, 1

++ and -- challenge

Try first. Then verify by compiling.

```
int x = 5;
int a = x++;
int b = ++x;

printf("%d %d %d", x, a, b);
```

Your answer

Write it here:

_____

Answer: 7 5 7

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Beginner Practice

1. Read two integers and print +, -, *, / and % results.
2. Read a number and determine whether it is even or odd using %.
3. Read two numbers and print the larger one using a relational operator.
4. Read age and print whether the person is eligible to vote.

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Medium Practice

1	Convert seconds into h:m:s using / and %.
2	Swap two integers using arithmetic operators only.
3	Read a 3-digit number and print its hundreds, tens and ones digits.
4	Check whether a number is divisible by both 3 and 5 using &&.
5	Find the maximum of three numbers using relational + logical operators.

Operator Mix Challenge

Write ONE program that:

- Reads two integers a and b.
- Prints sum, difference, product and remainder.
- Checks whether a > b.
- Checks whether both numbers are even.
- Prints the larger number using the ternary operator.

Bonus

Do the final maximum without if-else.

Operator Cheat Sheet

Arithmetic	`+ - * / %`
Relational	`== != > < >= <=`
Logical	`&& || !`
Assignment	`= += -= *= /= %=`
Increment	`++ --`
Conditional	`? :`
Bitwise	`& | ^ ~ << >>`

CONTEST TIP

Know the operator. Know the precedence. Test small values.

Practice Before Lecture 04(don’t need to submit)☺

1. Complete all 10 tasks from the task slides.
2. Create 5 of your own output-prediction questions.
3. Practice prefix vs postfix with at least 5 examples.
4. Try 5 bitwise examples using binary representation.
5. Explain why each answer is correct — don't just memorize.

LECTURE 03 COMPLETE

Operators are the building blocks of logic.

Practice → predict → compile → understand.